

DEMONSTRATE NUMERACY SKILLS

July/August 2024

SECTION A (40 MARKS)

Answer *all* the questions in this section

1. Evaluate $\frac{180 \div 3 - 150 \div 5}{4 \times 3 + 9 \times 2} - \frac{5 - 2 + 6}{4}$ (3 marks)
2. Given that the ratios $x: y = 2: 5$ and $y: z = 3: 4$, determine the ratio $x: y: z$ (3 marks)
3. The initial cost of a plot is Ksh. 800,000. At the end of each year, the land value increases by 2%.
Find the value of the plot at the end of three years. (3 marks)
4. Evaluate $\frac{(3^2)^{\frac{3}{2}} \times (8^2)^{\frac{1}{3}}}{2^{\frac{3}{2}} \times (3)^{\frac{1}{2}} - 1}$ (4 marks)
5. A picnic site is in form of a triangle ABC such that A and B are 120 m apart, angles $CAB = 80^\circ$ and $CBA = 40^\circ$. Determine its total area in hectares correct to 2 decimal places. (4 marks)
6. Solve the following simultaneous equation using elimination method. (4 marks)
$$3x + y = 7$$
$$6x + 3y = 12$$
7. A quantity y is inversely proportion to x^2 . Given that $y = 12$ when $x = 3$, find x when $y = 27$. (4 marks)
8. A straight line passes through the point (1, 5) and is perpendicular to the line $8y + 4x - 3 = 0$.
Find the equation of the line. (4 marks)
9. Three people Brian, Ian and Ryan contributed to a fund. Brian provided $\frac{2}{5}$ of the total, Ian gave $\frac{1}{3}$ of the remainder and Ryan provided Ksh. 8000. Determine the total amount contributed to the fund. (4 marks)
10. Given the formulae $m = \frac{ax^2}{x^2 + b} - c$, make x the subject and hence find the value of x when $c = 2$, $m = 3$, $b = 1$ and $a = 9$. (4 marks)

11. A washing basin is in shape of a hemisphere and has a capacity of 19.404 litres. Calculate its radius in centimetres. (take $\pi = \frac{22}{7}$) (3 marks)

SECTION B: (60 MARKS)

Answer any **three** questions in this section

12. (a) A conical flask of base radius 21 cm and height 35 cm is filled with water. Half of this water is then poured into a rectangular tank with a square base of side 12 cm. Find the depth of water in the tank in metres correct to 2 decimal places. (take $\pi = \frac{22}{7}$) (10 marks)

- (b) Table 1 shows the distribution of marks scored by 220 students in a numeracy skills test. **Table 1**

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of students	9	25	32	40	44	30	28	12

Determine the:

- (i) Mean (5 marks)
- (ii) Interquartile range (5 marks)
13. (a) Mr. Koech wishes to fence off his rectangular plot whose area is 1200 m^2 . One of the side is already catered for by a wall and he has a fencing wire of 100 metres long. Form a quadratic equation and solve it to find two possible dimensions of the plot. (10 marks)
- (b) A cyclist travels his first part of the journey at 25 km/h , and the remaining 70 km of the journey at 20 km/h . If the whole journey took 5 hours, find the:
- (i) Distance travelled at first part. (6 marks)
- (ii) Average speed of the journey. (4 marks)
14. (a) Three rangers Kamau, Peter and John were in patrol to lay an ambush for poachers. Peter was positioned 50 km on a bearing of 043° from Kamau, while John was placed on a bearing of 160° from Peter and 133° from Kamau. Sketch their position and determine the area of patrol to the nearest km^2 . (10 marks)

- (b) Three quantities P, Q and R are such that P varies directly as Q and inversely as the square of R. If $P=8$, $Q=16$ and $R=12$, determine the:

- (i) The equation connecting the three quantities (4 marks)
(ii) Percentage change in Q if P increases by 25% and C decreases by 5%. (6 marks)

15. (a) Table 2 shows marks scored by 170 students in a digital literacy test. **Table 2**

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	10	20	x	40	y	25	15

Given that the median mark is 35, determine the:

- (i) Values of x and y (5 marks)
(ii) Standard deviation of the marks. (5 marks)
- (b) An arc of length 8.8 m subtends an angle of 72° at the centre of a circle. Calculate the:
(Take $\pi = \frac{22}{7}$)
- (i) Radius of the circle. (4 marks)
(ii) Area of the segment bounded by the arc and the corresponding chord. (6 marks)

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